Chapter 2
Numerical Integrators

At its most basic level, molecular dynamics is about mapping out complicated point sets using trajectories of a system of ordinary differential equations (or, in Chaps. 6–8, a stochastic-differential equation system). The sets are typically defined as the collection of probable states for a certain system. In the case of Hamiltonian dynamics, they are directly associated to a region of the energy landscape. The trajectories are the means by which we efficiently explore the energy surface. In this chapter we address the design of numerical methods to calculate trajectories.

When we use the equations of motion for an atomic system and compute a trajectory, we are producing what we hope to be a representative path of the system. For the moment, we discount any external interactions (such as due to heating at a boundary or other driving forces), so we can think of the system as a closed, Hamiltonian dynamics model. This introduces the *microscopic* perspective which concerns the detailed description of the instantaneous atomic positions and velocities as time is varied. These microscopic paths are the cornerstone of statistical mechanical theory, which is the tool that we will eventually develop for understanding molecular systems at a more abstract level, thus it is essential in developing a computational methodology that we have an understanding of how to generate trajectories reliably and efficiently.

The challenge before us is to compute solutions of

$$\dot{\boldsymbol{q}} = \boldsymbol{M}^{-1}\boldsymbol{p}, \qquad \dot{\boldsymbol{p}} = \boldsymbol{F}(\boldsymbol{q}) = -\nabla U(\boldsymbol{q}),$$

or, more compactly, with $\boldsymbol{z}$ representing the collection of all positions and momenta,

$$\dot{\boldsymbol{z}} = \boldsymbol{f}(\boldsymbol{z}), \qquad \boldsymbol{f}(\boldsymbol{z}) = \boldsymbol{J}\nabla H, \tag{2.1}$$

where $\boldsymbol{J} = \begin{bmatrix} \boldsymbol{0} & \boldsymbol{I} \\ -\boldsymbol{I} & \boldsymbol{0} \end{bmatrix}$, and $H = \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2 + U(\boldsymbol{q})$ is the Hamiltonian.

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